Proceedings of the 3rd Conference on Fully Actuated System Theory and Applications
May 10-12, 2024, Shenzhen, China

An INS/GNSS Localization Method Based on Set-Membership Filter for Unmanned Aerial Vehicles

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Abstract: This paper studies a set-membership filter-based inertial integrated localization problem, where the inertial navigation system (INS) error and Global Navigation Satellite System (GNSS) measurement error are unknown-but-bounded (UBB) noises. To solve this problem, we establish a strapdown INS error model and a GNSS measurement model with UBB noises. Based on the optimal set-membership filtering framework, we propose a constrained zonotopic fast resurisive localization method. Compared with the existing probabilistic integrated navigation methods, the proposed method does not need to know the specific noise distribution and has better versatility and robustness. Finally, the numerical simulation results of the INS/GNSS navigation corroborate the effectiveness of this method.

Key Words: INS/GNSS, Constrained Zonotope, Set-Membership Filter

2024 3rd Conference on Fully Actuated System Theory and Applications (FASTA) | 979-8-3503-7369-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/FASTA61401.2024.10595218

1 Introduction

Nowadays, unmanned aerial vehicles (UAVs) have been widely used in various aspects of real life. However, the task of determining the position of a UAV poses many challenges. To localize themselves under unknown environments, UAVs extensively utilize the Global Navigation Satellite System (GNSS) and inertial navigation system (INS) integrated navigation to obtain high-precision navigation position information.

GNSS can provide high-precision three-dimensional positioning information, with accuracy reaching the centimeter level [1]. However, its performance is influenced by external environmental factors and satellite visibility, leading to situations where update frequency is low and signal interruptions occur in certain cases [2]. On the contrary, INS is an autonomous system that continuously integrates real angular rates and specific forces measured by gyroscopes and accelerometers to obtain position, velocity, and attitude information. However, gyro drift and accelerometer errors of the INS tend to increase over time [3].

Combining INS and GNSS can compensate for the individual shortcomings of each system, ultimately enhancing the precision of the navigation system. Traditionally, probabilistic approaches are employed for data fusion of these two sensors such as UKF [4] or EKF [5].

However, when GNSS measurement information deteriorates due to interference, and environmental vibrations affect the inertial navigation devices in complex high-dynamic environments, accurately representing their noise probability distribution becomes challenging. In such scenarios, these probabilistic methods have limitations [6]. Therefore, addressing combined navigation in the presence of unknown noise distributions of INS/GNSS sensors is crucial for enhancing the adaptability and robustness of integrated navigation systems.

Set-Membership Filter [7] as a set-based filtering theory, treats noise as unknown-but-bounded noise, and can effectively address situations where the noise distribution of INS and GNSS is unknown. In [8], a comprehensive set-based GNSS positioning method is proposed. This method named Lane Boundary Cooperative Augmented Set-membership GNSS Positioning. In [8], an approach to three-dimensional map-aided GNSS positioning method based on set-valued estimation is presented. The authors propose zonotope shadow matching (ZSM) as a set-valued three-dimensional map-aided GNSS positioning method. The strapdown INS error model is provided in [9], depicting how errors in the INS accumulate over time.

Building on existing work, we aim to develop a set-membership filter combined navigation method for INS/GNSS based on unknown-but-bounded noise description, with the purpose of estimating the UAV position. Our primary contributions are as follows:

- We first present the strapdown INS error model based on unknown-but-bounded errors. The errors in the inertial navigation accumulate over time in the form of unknown-but-bounded noise.
- Most importantly, within the set-membership filter framework, we propose an INS/GNSS integrated navigation method based on unknown-but-bounded noise. This approach allows for obtaining the UAV position, which is represented in a set-valued form.

The remainder of this paper is structured as follows. Section II introduces tools for describing set, Section III describes the notation and the system model, Section IV presents the method, Section V describes the experiment and results, Section VI is the conclusion.

2 Preliminaries

In this paper, constrained zonotopes are adopted to accurately describe a set's convex inclusion. They are used to describe the uncertainty range of the INS error. Constrained zonotopes are an extension of zonotopes, more specifically proposed in [10].

Definition 1 (Constrained Zonotope): A set $Z \subset \mathbb{R}^n$ is a constrained zonotope if there exists $(G_z, c_z, A_z, b_z) \in \mathbb{R}^{n\times n_g} \times \mathbb{R}^n \times \mathbb{R}^{n_c\times n_g} \times \mathbb{R}^{n_c}$ such that

$$Z = \{G_z\xi + c_z : \|\xi\|_\infty \leq 1, A_z\xi = b_z\}, \qquad (1)$$

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written as $Z(G_z, c_z, A_z, b_z)$.

Apart from being able to describe sets exactly, constrained zonotope is also closed with respect to basic operations on sets. More specifically, the operations are indicated in Proposition 1.

Proposition 1 (The basic set operations [10]): The constrained zonotope $Z = Z(G_z, c_z, A_z, b_z) \in \mathbb{R}^n$, $W = Z(G_w, c_w, A_w, b_w) \in \mathbb{R}^n$, $Y = Z(G_y, c_y, A_y, b_y) \in \mathbb{R}^k$ and $R \in \mathbb{R}^{k\times n}$.

$$
\begin{aligned}
&RZ = Z(RG_z, Rc_z, A_z, b_z)\\
&Z \oplus W = Z([G_z\ G_w], c_z + c_w, \begin{bmatrix} A_z & 0 \\ 0 & A_w \end{bmatrix}, \begin{bmatrix} b_z \\ b_w \end{bmatrix})\\
&Z \cap_R Y = Z([G_z\ 0], c_z, \begin{bmatrix} A_z & 0 \\ 0 & A_y \\ RG_z & -G_y \end{bmatrix}, \begin{bmatrix} b_z \\ b_y \\ c_y - Rc_z \end{bmatrix})
\end{aligned} \tag{2}
$$

In this work, the interval hull $\square Z$ of a constrained zonotope is computed using linear programming (LP).

Proposition 2 (Interval Hull): A constrained zonotope can be bounded by a interval hull as follow:

$$
\begin{aligned}
&Z(G, c, A, b) \subseteq \square Z(p, H)\\
&\begin{cases}
p = \prod_{i=1}^{n} \mathrm{mid}([\rho^{i-}, \rho^{i+}])\\
H = \prod_{i=1}^{n} \mathrm{diag}(rad([\rho^{i-}, \rho^{i+}]))\\
\rho^{i-} = \min\{G^i\xi + c^i \colon A\xi = b, \|\xi\|_\infty \leqslant 1\}\\
\rho^{i+} = \max\{G^i\xi + c^i \colon A\xi = b, \|\xi\|_\infty \leqslant 1\}
\end{cases}
\end{aligned} \tag{3}
$$

where $\mathrm{mid}(s,t) := \frac{1}{2}(s+t), \mathrm{rad}(s,t) := \frac{1}{2}|s-t|$.

In this work, we use uncertain variables to describe the variables:

Definition 2 (Uncertain Variables [7]): Unlike random variables, which can be described by probability distributions, uncertain variables lack any information about probability but can be described by their range of values.

$$
[\![\mathbf{x}]\!] := \{\mathbf{x}(\omega) : \omega \in \Omega\}, \tag{4}
$$

where $\mathbf{x}$ is referred to as an uncertain variable, and $[\![\mathbf{x}]\!]$ is referred to as the range of the uncertain variable.

3 System Model

In practical unmanned aerial systems, inertial navigation estimation is commonly employed to estimate the position, velocity, and attitude of the unmanned aerial vehicle. However, the errors in navigation estimation tend to increase over time. This paper proposes an analysis of the uncertain range of navigation estimation errors using uncertainty variables. The UAV state and inertial measurements [11] can be expressed by:

$$
\begin{bmatrix} \tilde{\phi} \\ \tilde{\nu}^n \\ \tilde{p} \end{bmatrix} = \begin{bmatrix} \phi \\ \nu^n \\ p \end{bmatrix} + \begin{bmatrix} \delta\phi \\ \delta\nu^n \\ \delta p \end{bmatrix}, \tag{5}
$$

$$
\begin{bmatrix} \tilde{\omega}_{ib}^b \\ \tilde{f}^b \end{bmatrix} = \begin{bmatrix} \omega_{ib}^b \\ f^b \end{bmatrix} + \begin{bmatrix} \delta\omega_{ib}^b \\ \delta f^b \end{bmatrix}, \tag{6}
$$

where $\tilde{\phi}$, $\tilde{\nu}^n$ and $\tilde{p}$ are output state of attitude and speed and position, ϕ, ν^n and p are the ground-truth state, $\delta\phi$, $\delta\nu^n$ and δp are state errors. $\tilde{\omega}_{ib}^b$ and $\tilde{f}^b$ are measurements of angular velocity and specific force, ω_{ib}^b and f^b are truth-value, $\delta\omega_{ib}^b$ and δf^b are errors. More specifically, the errors are indicated in Lemma 1.

Lemma 1 (unknown-but-bounded errors): The UAV state and IMU errors can be approximated by uncertain variable in Definition 2 as

$$
\begin{aligned}
&\delta\phi = [\delta\theta, \delta\gamma, \delta\varphi] \in [\![\delta\boldsymbol{\phi}]\!],\\
&\delta\nu^n = [\delta\nu_E, \delta\nu_N, \delta\nu_U] \in [\![\delta\boldsymbol{\nu}^n]\!],\\
&\delta p = [\delta L, \delta\lambda, \delta h] \in [\![\delta\boldsymbol{p}]\!],\\
&\delta\omega_{ib}^b = [\delta\omega_{ibx}^b, \delta\omega_{iby}^b, \delta\omega_{ibz}^b] \in [\![\delta\boldsymbol{\omega}_{ib}^b]\!],\\
&\delta f^b = [\delta f_x^b, \delta f_y^b, \delta f_z^b] \in [\![\delta\boldsymbol{f}^b]\!].
\end{aligned} \tag{7}
$$

Remark 1: It is usually assumed that the noise belongs to a Gaussian distribution in previous methods [11]. In fact, the error distribution of the UAV state and IMU is difficult to be described by a deterministic distribution. However, based on UAV parameters and IMU sensor information and statistics, the magnitude of the error regions can be obtained.

To establish the INS/GNSS integrated navigation model, we choose the strapdown INS error model [12]. The functional relation is given in Lemma 2.

Lemma 2 (Strapdown INS error model): The strapdown INS error model for $\boldsymbol{x} = [\delta\boldsymbol{\phi}, \delta\boldsymbol{\nu}^n, \delta\boldsymbol{p}]$ and $\boldsymbol{w} = [\delta\boldsymbol{\omega}_{ib}^b, \delta\boldsymbol{f}^b]$ are as follows:

$$
\dot{\boldsymbol{x}} = F\boldsymbol{x} + G\boldsymbol{w}, \tag{8}
$$

where

$$
F = \begin{bmatrix} M_{aa} & M_{a\nu} & M_{ap} \\ M_{\nu a} & M_{\nu\nu} & M_{\nu p} \\ M_{pa} & M_{p\nu} & M_{pp} \end{bmatrix}, G = \begin{bmatrix} -C_b^n & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & C_b^n \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix},
$$

and the details can be found in Appendix A.

By discretizing the strapdown INS error model (8), one can obtain the discrete-time model

$$
\boldsymbol{x}_{k+1} = A_k\boldsymbol{x}_k + B_k\boldsymbol{w}_k, \tag{9}
$$

where $A_k = e^{F\Delta t}$ and $B_k = \int_0^{F\Delta t} e^{F\tau} d\tau G$; Δt denotes the time duration from t_{k-1} to t_k.

Equation (9) represents the uncertainty error range of the inertial navigation estimation results. In order to correct this accumulating error, unmanned aerial vehicles require additional measurement information. This paper considers GNSS measurement information. However, measurement information also entails an error range, which we describe and analyze in the form of uncertainty variables. The GNSS measurement error is given by:

$$
z_k = \tilde{p}_{INS}^n - \tilde{p}_{GNSS}^n = H_k\boldsymbol{x}_k + \boldsymbol{v}_k, \tag{10}
$$

where

$$
H_k = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \boldsymbol{I}_{3\times3} \end{bmatrix}.
$$

H_k represents the measurement matrix, z_k represents the measurement output, $\tilde{p}_{INS}^n$ denotes the estimation results of the inertial navigation system, $\tilde{p}_{GNSS}^n$ denotes the estimation results of the Global Navigation Satellite System, and $\boldsymbol{v}_k$ represents the uncertainty of measurement errors.

To achieve more accurate estimation of UAV states, it is necessary to fuse the system error uncertainty and measurement error uncertainty in Equation (9) and Equation (10)

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through filtering, in order to eliminate the cumulative errors of the Inertial Navigation System. For the uncertainty variable description of error uncertainty filtering, this paper designs a combined navigation method based on the optimal set membership filtering framework.

For any $k \geqslant 0$, let $\hat{\boldsymbol{X}}_k$ denote the set of all states $[\![\boldsymbol{x}_k]\!] \in \hat{\boldsymbol{X}}_k$ that are consistent with (9) the discrete time model, the measured output sequence up to time k, $(z_0, z_1, \ldots, z_k)$, and the unknownbut-bounded uncertain state $[\![\boldsymbol{x}_0]\!] \subseteq \hat{\boldsymbol{X}}_0$, $[\![\boldsymbol{w}_k]\!] \subseteq \boldsymbol{W}_k$ and $[\![\boldsymbol{v}_k]\!] \subseteq \boldsymbol{V}_k$. $\hat{\boldsymbol{X}}_k$ is given recursively by the SMFing is indicated in Lemma 2.

Lemma 3 (SMFing framework): Based on the optimal SMFing framework [7], the general scheme is given by the following recursion:

- **Initialization.** Set the initial prior set $[\![\boldsymbol{x}_0]\!] \subseteq \hat{\boldsymbol{X}}_0$.
- **Prediction.** For $k \geqslant 0$, given $\hat{\boldsymbol{X}}_k$, the prior set $\bar{\boldsymbol{X}}_{k+1}$ is predicted by the equation (9) is

$$\bar{\boldsymbol{X}}_{k+1} \supseteq \{A_k x_k + B_k w_k \colon x_k \in [\![\boldsymbol{x}_k]\!], w_k \in [\![\boldsymbol{w}_k]\!]\}. \tag{11}$$

- **Update.** For $k \geqslant 0$, given the measured output z_{k+1}, the posterior set $\hat{\boldsymbol{X}}_{k+1}$ is

$$\hat{\boldsymbol{X}}_{k+1} \supseteq \{x_{k+1} \in \bar{\boldsymbol{X}}_{k+1} \colon H_{k+1}x_{k+1} + v_{k+1} = z_{k+1}, v_{k+1} \in [\![\boldsymbol{v}_k]\!]\}. \tag{12}$$

The new estimated state $[\![\boldsymbol{x}_{k+1}]\!] \subseteq \hat{\boldsymbol{X}}_{k+1}$.

To ensure the optimality of the filtering, this paper makes the following assumptions:

Assumption 1 (Unrelated Noises and Initial State): $\forall k \in \mathbb{N}_0$, $\boldsymbol{w}_k, \boldsymbol{v}_k, \boldsymbol{x}_0$ are unrelated.

In the remainder of the paper, our goal is to compute these two steps (11)-(12) using SMFing.

4 Method

The combined INS/GNSS navigation method with the set-membership filter of IMU data is derived in this section, and the process is shown in Fig. 1.

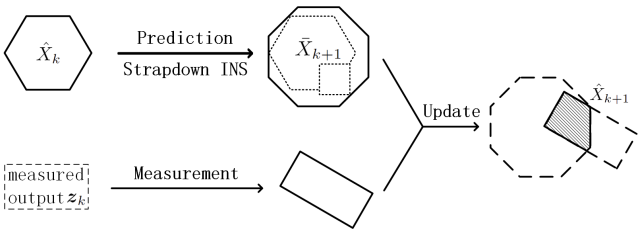


Fig. 1: Procedure of SMF Method.

$[\![\boldsymbol{w}_k]\!] \subseteq \boldsymbol{W}_k$ and $[\![\boldsymbol{v}_k]\!] \subseteq \boldsymbol{V}_k$ are respectively the state and measurement error which satisfy the following equations:

$$[\![\boldsymbol{w}_k]\!] \subseteq \boldsymbol{W}_k = Z(\hat{G}_k^w, \hat{c}_k^w, \hat{A}_k^w, \hat{b}_k^w), \tag{13}$$

$$[\![\boldsymbol{v}_k]\!] \subseteq \boldsymbol{V}_k = Z(\hat{G}_k^v, \hat{c}_k^v, \hat{A}_k^v, \hat{b}_k^v). \tag{14}$$

The initial state of system satisfies the equation (9) and Lemma 3:

$$[\![\boldsymbol{x}_0]\!] \subseteq \hat{\boldsymbol{X}}_0 = Z(\hat{G}_0^x, \hat{c}_0^x, \hat{A}_0^x, \hat{b}_0^x). \tag{15}$$

On the basis of the INS error discrete time model, the prediction system's state satisfy the following equations:

$$\bar{\boldsymbol{X}}_{k+1} = A_k \hat{\boldsymbol{X}}_k \oplus B_k \boldsymbol{W}_k. \tag{16}$$

Equation (16) demonstrates that prediction state set, $A_k \in \mathbb{R}$ and $B_k \in \mathbb{R}$ are approximation matrices.

The specific calculation of A_k and B_k are as follows:

$$A_k = e^{F\Delta t} = \sum_{i=0}^{\infty} \frac{(F\Delta t)^i}{i!} \approx \boldsymbol{I} + F\Delta t + \frac{1}{2}(F\Delta t)^2, \tag{17}$$

$$B_k = \int_0^{F\Delta t} e^{F\tau} d\tau G = \sum_{i=0}^{\infty} \frac{\left(F^i \Delta t^{i+1}\right)}{(i+1)!} G \approx (\Delta t + \frac{1}{2}F\Delta t^2 + \frac{1}{6}\Delta t^3 F^2)G. \tag{18}$$

Finally, the prediction system's state as following equation:

$$\bar{\boldsymbol{X}}_{k+1} = (1 + F\Delta t + \tfrac{1}{2}(F\Delta t)^2) Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x) \oplus ((\Delta t + \tfrac{1}{2}F\Delta t^2 + \tfrac{1}{6}\Delta t^3 F^2)G) Z(\hat{G}_k^w, \hat{c}_k^w, \hat{A}_k^w, \hat{b}_k^w). \tag{19}$$

The function of update step is to revise the prediction set $\bar{\boldsymbol{X}}_{k+1}$ by measured output (10). Measurement set $\boldsymbol{Y}$ is also represented by constrained zonotope like the prediction set, the measurement uncertain set is represented by $Z(\hat{G}^y, \hat{c}^y, \hat{A}^y, \hat{b}^y)$.

Then the estimated state set $\hat{\boldsymbol{X}}_{k+1}$ given by the following equation:

$$\{x_{k+1} \in \bar{\boldsymbol{X}}_{k+1} : H_{k+1}x_{k+1} + v_{k+1} = z_{k+1}, v_{k+1} \in \boldsymbol{V}_{k+1}\} \subseteq \hat{\boldsymbol{X}}_{k+1} = \bar{\boldsymbol{X}}_{k+1} \cap_R \boldsymbol{Y}_{k+1}, \tag{20}$$

where $R = H_{k+1}$, and $\boldsymbol{Y}_{k+1} = z_k \oplus \boldsymbol{V}_{k+1}$.

Through Equation (20), we can obtain the estimation result of one iteration. However, as the number of iterations increases, the dimension of constrained zonotope will continuously increase, leading to a decrease in computational efficiency.

To handle this problem, we propose a fast iteration method (see Algorithm 1) based on OIT heuristic in [13], and we provide a line by line explanation as follows:

Algorithm 1 Fast Recursive Method

1: **Initialization:**Bounded $[\![\boldsymbol{x}_0]\!] \subseteq \hat{\boldsymbol{X}}_0 = Z(\hat{G}_0^x, \hat{c}_0^x, \hat{A}_0^x, \hat{b}_0^x)$, select slide window size δ.
2: **loop**
3: $\quad \square Z(\hat{p}_k, \hat{H}_k) \leftarrow Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x)$ by (3);
4: $\quad$ **if** $k \leqslant \delta$ **then**
5: $\quad\quad Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x) \leftarrow \hat{X}_{k-1}$ by (19) and (20);
6: $\quad$ **else if** $k > \delta$ **then**
7: $\quad\quad Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x) \leftarrow \square Z(\hat{p}_{k-\delta-1}, \hat{H}_{k-\delta-1})$ by (19) and (20);
8: $\quad$ **end if**
9: $\quad \hat{\boldsymbol{X}}_k = Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x)$;
10: $\quad k++$
11: **end loop**

- Line 1 is the initialization of Algorithm 1, where δ represents the chosen window size of Algorithm 1. The

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dimension of $\hat{\boldsymbol{X}}_k$ will be constrained based on the size of δ, with a larger δ resulting in more accurate results and a smaller δ leading to faster algorithm execution.

- Line 3 involves storing the estimation results of each iteration in Interval Hull.
- Line 5 is for $k \leqslant \delta$, which is the same as the classical constrained zonotope SMF algorithm.
- Line 7 is for $k > \delta$, which involves obtaining the estimation result for $Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x)$ after k iterations using $\square Z(\hat{p}_{k-\delta-1}, \hat{H}_{k-\delta-1})$ as the initial state over the time window $[k-\delta, k]$. It is a simplified version of the OIT inspired constrained zonotopic SMF in [13].

5 Simulation and Result

In the experimental study, we respectively use the position and velocity and attitude errors of UAV as the state and the angular velocity and acceleration errors were simulated in the time scale model of the inertial measuring unit, using the equations

$$\boldsymbol{x}_k \in \hat{\boldsymbol{X}}_k = \left[\ \delta\hat{\boldsymbol{\phi}}_k, \delta\hat{\boldsymbol{\nu}}_k^n, \delta\hat{\boldsymbol{p}}_k\ \right] = Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x), \tag{21}$$

$$\boldsymbol{w}_k \in \boldsymbol{W}_k = [\delta\boldsymbol{\omega}_{ib,k}^b, \delta\boldsymbol{f}_k^b] = Z(G_k^w, c_k^w, A_k^w, b_k^w). \tag{22}$$

IMU and their characteristics for clarity are given in the Table 1 below.

Table 1: Sensor Parameters of The Simulation

Specification	Value
Gyroscope constant bias (deg/h)	0.03
Accelerometer constant bias (ug)	100
Angular random walk	0.001
Velocity random walk	5
GNSS error(m)	10
IMU rate (Hz)	10
GNSS rate(Hz)	10
Simulation step size	540

From Fig. 2 which displays the interval estimation results for the unmanned aerial vehicle's position. The interval estimation represents the range of estimates for each component derived from the ensemble filtering estimation. Fig. 3 shows the position estimation results of the UAV in the simulation. The point estimation result is obtained as the median of the interval estimation result derived from the ensemble filtering position estimation.

6 Conclusion

In this paper, combining with the INS and GNSS by Set-Membership Filter. Aiming at the situation that the error distribution of INS and GNSS is unknown in the actual system, the bound of unknown-but-bounded noise inclusion estimation is used, which is a powerful supplement to the INS integrated navigation method based on probability. Apart from inertial integrated navigation with GNSS as measurement, there are other more practical integrated navigation methods, such as visual-inertial odometry (VIO). Therefore, we will carry out in-depth research on SMF-based, more reliable and more general integrated navigation methods in the future. In future work, we will address the role of the INS linearization error in the SMF prediction equation.

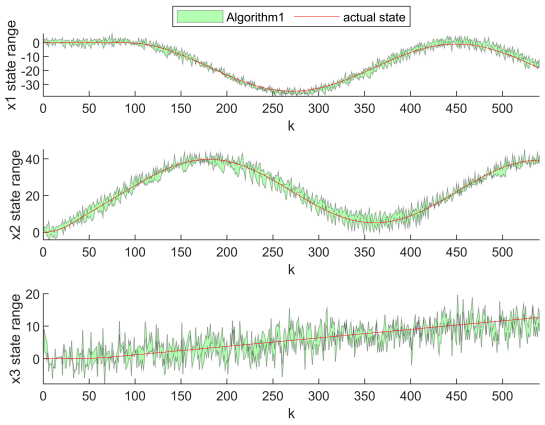


Fig. 2: The interval estimation results of state.The results indicate that the proposed Algorithm 1 is able to enclose the true position state of the unmanned aerial vehicle.

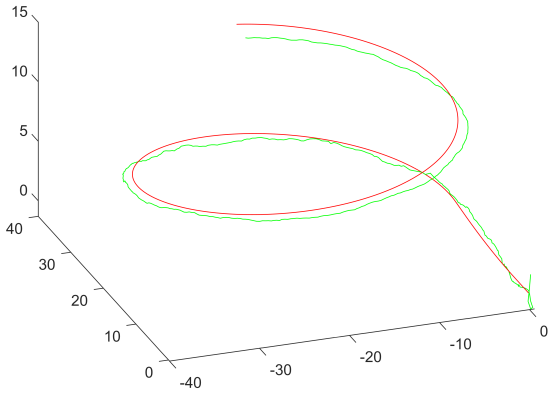


Fig. 3: The point estimation results of state.The red line is the true state and the green line is the estimated state.

A INS Error propagation formula

The following content demonstrates the specific form of F in the strapdown INS error model. i represents the inertial coordinate system; e represents the earth coordinate system; n represents the navigation coordinate system; b represents the body coordinate system. R_M and R_M represent the Earth's radii parameters.

$$M_{aa} \triangleq -\left(\boldsymbol{\omega}_{\boldsymbol{in}}^{\boldsymbol{n}}\times\right):$$

$$\begin{cases} M_{aa}(1,1) = M_{aa}(2,2) = M_{aa}(3,3) = 0 \\ M_{aa}(1,2) = \omega_{ie}\sin L + \dfrac{\nu_E}{R_N+h}\tan L \\ M_{aa}(1,3) = -\omega_{ie}\cos L - \dfrac{\nu_E}{R_N+h} \\ M_{aa}(2,1) = -\omega_{ie}\sin L - \dfrac{\nu_E}{R_N+h}\tan L \\ M_{aa}(2,3) = -\dfrac{\nu_N}{R_M+h} \\ M_{aa}(3,1) = \omega_{ie}\cos L + \dfrac{\nu_E}{R_N+h} \\ M_{aa}(3,2) = \dfrac{\nu_N}{R_M+h}. \end{cases}$$

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$$M_{a\nu}=\begin{bmatrix} 0 & -\dfrac{1}{R_M+h} & 0 \\ \dfrac{1}{R_N+h} & 0 & 0 \\ \dfrac{\tan L}{R_N+h} & 0 & 0 \end{bmatrix},$$

$$M_{ap}=\begin{bmatrix} 0 & 0 & \frac{\nu_N}{(R_M+h)^2} \\ -\omega_{ie}\sin L & 0 & -\frac{\nu_E}{(R_N+h)^2} \\ \omega_{ie}\cos L+\frac{\nu_E\sec^2 L}{R_N+h} & 0 & -\frac{\nu_E\tan L}{(R_N+h)^2} \end{bmatrix},$$

$$M_{\nu a}\triangleq(\boldsymbol{f}^n\times)=\begin{bmatrix} 0 & -f_U^n & f_N^n \\ f_U^n & 0 & -f_E^n \\ -f_N^n & f_E^n & 0 \end{bmatrix},$$

$M_{\nu\nu}$:

$$\begin{cases} M_{\nu\nu}(1,1)=\dfrac{\nu_N\tan L-\nu_U}{R_N+h} \\ M_{\nu\nu}(1,2)=2\omega_{ie}\sin L+\dfrac{\nu_E\tan L}{R_N+h} \\ M_{\nu\nu}(1,3)=-2\omega_{ie}\cos L-\dfrac{\nu_E}{R_N+h} \\ M_{\nu\nu}(2,1)=-2\omega_{ie}\sin L-\dfrac{2\nu_E\tan L}{R_N+h} \\ M_{\nu\nu}(2,2)=-\dfrac{\nu_U}{R_M+h} \\ M_{\nu\nu}(2,3)=-\dfrac{\nu_N}{R_M+h} \\ M_{\nu\nu}(3,1)=2\omega_{ie}\cos L+\dfrac{2\nu_E}{R_N+h} \\ M_{\nu\nu}(3,2)=\dfrac{2\nu_N}{R_M+h} \\ M_{\nu\nu}(3,3)=0 \end{cases}$$

$M_{\nu p}$:

$$\begin{cases} M_{\nu p}(1,1)=\dfrac{\nu_E\nu_N\sec^2 L}{R_N+h}+2\nu_N\omega_{ie}\cos L+2\nu_U\omega_{ie}\sin L \\ M_{\nu p}(1,2)=0 \\ M_{\nu p}(1,3)=\dfrac{\nu_E\nu_U-\nu_E\nu_N\tan L}{(R_N+h)^2} \\ M_{\nu p}(2,1)=-2\nu_E\omega_{ie}\cos L-\dfrac{\nu_E^2\sec^2 L}{R_N+h} \\ M_{\nu p}(2,2)=0 \\ M_{\nu p}(2,3)=\dfrac{\nu_E^2\tan L}{(R_N+h)^2}+\dfrac{\nu_N\nu_U}{(R_M+h)^2} \\ M_{\nu p}(3,1)=-2\nu_E\omega_{ie}\sin L \\ M_{\nu p}(3,2)=0 \\ M_{\nu p}(3,3)=-\dfrac{\nu_E^2}{(R_N+h)^2}-\dfrac{\nu_N^2}{(R_M+h)^2} \end{cases}$$

$$M_{pa}=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$M_{p\nu}=\begin{bmatrix} 0 & \frac{1}{R_M+h} & 0 \\ \frac{\sec L}{R_N+h} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$M_{pp}=\begin{bmatrix} 0 & 0 & -\frac{\nu_N}{(R_M+h)^2} \\ \frac{\nu_E\tan L\sec L}{R_N+h} & 0 & -\frac{\nu_E\sec L}{(R_N+h)^2} \\ 0 & 0 & 0 \end{bmatrix}$$

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